

GLASGOW COLLEGE UESTC

Final Exam paper

Electromagnetic Field and Microwave Technology (4002)

Date: 18th June 2022

Time: 14:30-16:30

Attempt all PARTS. Total 100 marks

Use one answer sheet for each of the questions in this exam.

Show all work on the answer sheet.

Make sure that your University of Glasgow and UESTC Student Identification Numbers are on all answer sheets.

An electronic calculator may be used provided that it does not allow text storage or display, or graphical display.

All graphs should be clearly labelled and sufficiently large so that all elements are easy to read.

The numbers in square brackets in the right-hand margin indicate the marks allotted to the part of the question against which the mark is shown. These marks are for guidance only.

DATA/FORMULAE SHEET IS PROVIDED AT THE END OF PAPER

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- Q1 (a) When the circuit theory is not valid, the theory of electromagnetic field need to be applied. State the difference between the theory of electromagnetic field and circuit theory, focus on the parameters of characteristic when you study on these two theories. [3]
- (b) Define and explain the meaning of scalar potential in static electric field and time-varying electromagnetic field. [3]
- (c) The interface between two media is $x=0$ plane, the magnetic fields on the interface are $\vec{H}_1 = 2\vec{e}_x + 4\vec{e}_y + 3\vec{e}_z$ ($x = 0_+$) and $\vec{H}_2 = -3\vec{e}_x + 2\vec{e}_y - 5\vec{e}_z$ ($x = 0_-$), what is the surface current density $\vec{J}_s$ on the interface. [7]
- (d) There are three types of polarization: linear polarization, circular polarization and elliptical polarization,
- (i) Explain the difference between them. [5]
- (ii) Illustrate the type of the polarization for the following uniform plane wave. $\vec{E}(z) = \vec{e}_x jE_m e^{jkz} + \vec{e}_y jE_m e^{jkz}$; $\vec{E}(z) = \vec{e}_x E_m e^{-jkz} - \vec{e}_y jE_m e^{-jkz}$ [7]

Q2 An infinite long line current I (along z axis) is located on the interface (xoz plane) of two semi-infinite magnetic media, as shown in **Figure Q1**. For $y>0$, the permeability is μ_1 , for $y<0$, the permeability is μ_2 . Find:

- (a) The boundary condition used on the interface. [4]
- (b) The magnetic flux density $\mathbf{B}$ and magnetic field intensity $\mathbf{H}$ and magnetization vector $\mathbf{M}$. [12]
- (c) The distribution of magnetized surface current density on the interface. [3]
- (d) The magnetized current close to the line current I . [6]

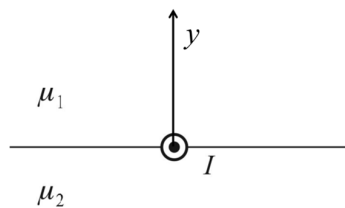


Figure Q1

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- Q3 (a) Explain the definition and meaning of displacement current? Does it denote that electric field is the source of magnetic field? [3]
- (b) What is Poynting vector? State the physical meaning of the direction of Poynting vector. [3]
- (c) Explain the characteristics of circular polarized TEM wave. [4]
- (d) Assume the electric field intensity of a uniform plane wave is $\vec{E} = (\vec{e}_x 3 + \vec{e}_y 4)e^{-\frac{\pi}{3}j(4x+Ay+\sqrt{11}z)}$, find.
- (i) The value of A. [5]
- (ii) The wavelength of the uniform plane wave. [5]
- (iii) The polarization of the uniform plane wave: [5]

- Q4 A uniform plane wave is normally incident from air to a perfect conductor at $z=0$, as shown in **Figure Q2**. The complex magnetic field is $\vec{H}_i(z) = H_0(2j\vec{e}_x + \vec{e}_y)e^{-j2\pi z}$, find:
- (a) The corresponding incident electric field. [5]
- (b) The total electromagnetic field in air. [10]
- (c) The positions of the nodes and antinode for the total electric field. [5]
- (d) The current density on the surface of conductor. [5]

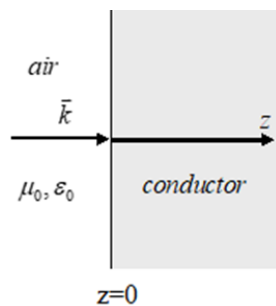


Figure Q2

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Appendix

Cylindrical coordinate system

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla^2 u = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2}$$

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{e}_\rho & \rho \mathbf{e}_\phi & \mathbf{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

Spherical coordinate system

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{e}_r & r \mathbf{e}_\theta & r \sin \theta \mathbf{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

$$\nabla^2 u = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial u}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}$$

End of question paper